Corrections (red). Wilson-line orientation: every U^{AB} has A = colour after the shock and B = colour before it, so ρ always sits on the second index and $U^{\dagger bc} = U^{cb}$. The red indices were fixed by imposing this rule term by term; the rule has a unique solution in every term.

JIMWLK evolution terms from genuine three ρ's

1 Group I (with c.c.)

1.1 Row I

Only 4 ρ's

1.2 Row II

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\left[-\bar{\mathbb{A}}_i^{\dagger\,(1)\,a}(k^+,-w') + \mathbb{A}_i^{\dagger\,(1)\,b'}(k^+,-w')U^{b'a}(w')\right]\times \left[U^{ab}(w)\mathbb{A}_i^{(3)\,b}(k^+,-w) - \bar{\mathbb{A}}_i^{(3)\,a}(k^+,-w)\right]$$

$$\begin{aligned}\frac{d^3N_{2\,a}}{d^3k} =& -\frac{1}{(2\pi)^3}\frac{i\,g^4\,f^{hbc}}{32\pi^5}\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,x',z}\frac{(x'-w')^i}{(x'-w')^2}\left[\frac{(x-w)^i}{(x-w)^2}\frac{(w-z)\cdot(y-z)}{(w-z)^2\,(y-z)^2}\right.\\ &\left.-\frac{(y-w)^i}{(y-w)^2}\frac{(w-z)\cdot(x-z)}{(w-z)^2\,(x-z)^2}\right]\times\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{ah}(w)\,\{\rho^b(x),\rho^c(y)\}\right]\\ &-\frac{1}{(2\pi)^3}\frac{i\,g^4\,f^{abc}}{32\pi^4}\frac{1}{k^+}\log\left(\frac{k^+}{\Lambda}\right)\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,x',z}\frac{(x'-w')^i}{(x'-w')^2}\left[\frac{(x-w)^i}{(x-w)^2}\log\frac{(w-y)^2}{(x-y)^2}\right.\\ &\left.-\frac{(y-w)^i}{(y-w)^2}\log\frac{(w-x)^2}{(x-y)^2}\right]\times\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{bd}(x)\,U^{ce}(y)\,\{\rho^d(x),\rho^e(y)\}\right]\end{aligned} \tag{1.1}$$

1.3 Row III

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\int\frac{dp^+}{2\pi}\int_z\left[-\bar{\mathbb{A}}_i^{\dagger\,(1)\,a}(k^+,-w') + \mathbb{A}_i^{\dagger\,(1)\,b'}(k^+,-w')U^{b'a}(w')\right]\times \left[-\bar{\mathbb{B}}_{3ji}^{\dagger\,ba}(k^+,w;p^+,-z)\bar{\mathbb{A}}_j^{(1)\,b}(p^+,-z) + \bar{\mathbb{B}}_{3ji}^{\dagger\,ba}(k^+,w;p^+,-z)U^{bc}(z)\mathbb{A}_j^{(1)\,c}(p^+,-z)\right]$$

We write it in two parts:

$$\begin{aligned}\frac{d^3N_{1\,\rm NLO}}{d^3k} =& \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\left\{\right.\\ &\frac{ig^4\,f^{abd}}{16\pi^5}\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2\,(x-w)^2\,(x-z)^2\,(y-z)^2}\\ &\left.\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right]\right\}\end{aligned} \tag{1.2}$$

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)} \Bigg\{ & \\
& - \frac{ig^4\, f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x'} \frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& + \frac{ig^4\, f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x'} \frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& - \frac{ig^4\, f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x'} \frac{(x'-w')\cdot(y-w)\,(x-z)\cdot(w-z)}{(x'-w')^2\,(y-w)^2\,(x-z)^2\,(w-z)^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(w)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& - \frac{ig^4\, f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x'} \frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& + \frac{ig^4\, f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x'} \frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \Bigg\}
\end{aligned}
\tag{1.3}
$$

We write it as follows:

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4\, f^{abd}}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,w,w',x'} & e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z) \\
\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times & \Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]
\end{aligned}
\tag{1.4}
$$

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{ig^4\, f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,w,w',x'} & e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z) \\
\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times & \Big[U^{de}(x)\,U^{bc}(w)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]
\end{aligned}
\tag{1.5}
$$

Corrections (Row III). (i) In the five-term list above, the signs of the first two terms (the $[(x-w)^2-(x-z)^2]^{-2}$ structures) have been corrected to $-,+$: they originate from N_2 of the Group-I file, which enters with an overall minus sign, so the four such structures cancel pairwise against the two coming from N_{3a} (terms four and five), leaving only the third term. (ii) In the rewritten N_2 the first Wilson line of the second bracket has been corrected to $U^{bc}(w)$ (it read $U^{bc}(z)$): only with $U^{bc}(w)$ does the rewrite agree with the term it descends from, and the collected Row-III block below. *Verified symbolically for arbitrary N_c*: with these corrections, $[K_2\text{-term of } N_1] + [N_2 \text{ as rewritten}] + \text{h.c.}$ equals the collected Group-I Row-III block exactly.

1.4 Row IV

$$
\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{2}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\int\frac{dp^+}{2\pi}\int_z & \Big[-\bar{\mathbb{A}}_i^{\dagger\,(1)\,a}(k^+,-w') + \mathbb{A}_i^{\dagger\,(1)\,b'}(k^+,-w')U^{b'a}(w')\Big]\times \\
& \Big[U^{ac}(w)U^{bd}(z)\bar{\mathbb{A}}_j^{\dagger\,(1)\,b}(p^+,-z)\mathbb{B}_{2ji}^{dc}(k^+,-w;p^+,-z) - \bar{\mathbb{A}}_j^{\dagger\,(1)\,b}(p^+,-z)\bar{\mathbb{B}}_{2ji}^{ba}(k^+,-w;\,p^+,-z)\Big]
\end{aligned}
$$

$$
\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} & e^{-ik\cdot(w'-w)}\,\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\left[\frac{(x-w)\cdot(x'-w')\,(z-w)\cdot(y-z)}{(z-w)^2\,(x-w)^2\,(x'-w')^2\,(y-z)^2}\right. \\
& \left. + \frac{(x-z)\cdot(y-z)\,(x-w)\cdot(x'-w')}{2\,(x-z)^2\,(x-w)^2\,(x'-w')^2\,(y-z)^2}\right]\times\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times \\
& \Big[f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^e(y)\rho^{c'}(x) - f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^e(y)\rho^d(x)\Big]
\end{aligned}
\tag{1.6}
$$

We can write it as follows:

$$
\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} & e^{-ik\cdot(w'-w)}\,\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\left[-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\right. \\
& \left. + \frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\right]\times\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times \\
& \Big[f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^e(y)\rho^{c'}(x) - f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^e(y)\rho^d(x)\Big]
\end{aligned}
\tag{1.7}
$$

1.5 Row V

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{2}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)} \int \frac{dp^+}{2\pi} \int_{x,z} \Big[- \bar{\mathbb{A}}_i^{\dagger\,(1)\,a}(k^+,-w') + \mathbb{A}_i^{\dagger\,(1)\,b'}(k^+,-w') U^{b'a}(w') \Big] \times$$
$$\Big[- \bar{\mathbb{C}}^{\dagger\,bca}_{1jki}(k^+, w-z; p^+, z-x) \bar{\mathbb{B}}^{bc}_{2jk}(k^+-p^+, -z; p^+, -x) + \bar{\mathbb{C}}^{\dagger\,dca}_{1jki}(k^+, w-z; p^+, z-x)\, U^{db}(x)\, U^{ce}(z)$$
$$\mathbb{B}^{be}_{2jk}(k^+-p^+, -z; p^+, -x) \Big]$$

$$\frac{d^3N_{1\,\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\int_{z,w,y,x',y',w'} e^{-ik\cdot(w'-w)} \frac{ig^4}{32\pi^5} \frac{1}{k^+} \log\left[\frac{k^+}{\Lambda}\right] \frac{(x'-w')\cdot(y-w)\,(w-z)\cdot(y'-z)}{(x'-w')^2\,(w-z)^2\,(y-w)^2\,(y'-z)^2}$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[f^{adc}\,U^{db}(z)\,U^{ce}(w)\,\{\rho^e(y),\rho^b(y')\} - f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\Big]$$
$$- \frac{1}{(2\pi)^3}\int_{w,z,y,x',y',w'} e^{-ik\cdot(w'-w)} \frac{ig^4}{32\pi^5} \frac{1}{k^+} \log\left[\frac{k^+}{\Lambda}\right] \frac{(x'-w')\cdot(y'-w)\,(z-w)\cdot(y-z)}{(x'-w')^2\,(z-w)^2\,(y-z)^2\,(y'-w)^2}$$
$$\times \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[f^{adc}\,U^{db}(w)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\} - f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\Big] \tag{1.8}$$

We can write this as (after some simplification)

$$\frac{d^3N_{1\,\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{ig^4}{16\pi^5}\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)} \Big[$$
$$- \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^h(x)\,\rho^e(y)\,U^{ad}(x')\,U^{be}(z)\,U^{ch}(w)$$
$$+ \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^h(x)\,\rho^e(y)\,U^{ad}(w')\,U^{be}(z)\,U^{ch}(w)$$
$$+ \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(y)$$
$$- \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(y)\Big] \tag{1.9}$$

$$\frac{d^3N_{2\,\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{ig^4}{16\pi^5}\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)} \Big[$$
$$\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^h(y)\,\rho^e(x)\,U^{ad}(x')\,U^{be}(w)\,U^{ch}(z)$$
$$- \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{ce}(y)\,U^{bh}(x)$$
$$- \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^h(y)\,\rho^e(x)\,U^{ad}(w')\,U^{be}(w)\,U^{ch}(z)$$
$$+ \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{ce}(y)\,U^{bh}(x)\Big] \tag{1.10}$$

2 Group II

2.1 Row I

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{4}{(2\pi)^3}\int_{\Lambda}^{\vee-k^+} \frac{dp^+}{2\pi}\int_{z,w,w'} e^{-ik(w'-w)} \Big[\mathbb{B}^{\dagger\,bc'}_{2ji}(k^+,-w';p^+,-z)U^{c'a}(w') - \bar{\mathbb{B}}^{\dagger\,c'a}_{2ji}(k^+,-w';p^+,-z)U^{bc'}(z)\Big]\times$$
$$\Big[U^{ac}(w)\mathbb{B}^{bc}_{2ji}(k^+,-w;p^+,-z) - U^{\dagger\,bc}(z)\bar{\mathbb{B}}^{ca}_{2ji}(k^+,-w;p^+,-z)\Big]$$

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik(w'-w)}\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\left[\frac{(x-w)\cdot(x'-w')\,(z-w)\cdot(y-z)}{(z-w)^2\,(x-w)^2\,(x'-w')^2\,(y-z)^2}\right.$$
$$\left.+\,\frac{(x-z)\cdot(y-z)\,(x-w)\cdot(x'-w')}{2\,(x-z)^2\,(x-w)^2\,(x'-w')^2\,(y-z)^2}\right]\Big[U^{ac'}(w')\,\{\rho^{c'}(x'),\rho^b(y)\}-U^{bc'}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,\{\rho^{d'}(x'),\rho^{e'}(y)\}\Big]\times$$
$$\Big[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\Big]$$
$$+\,\frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik(w'-w)}\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\left[\frac{(x-w')\cdot(x'-w)\,(z-w')\cdot(y-z)}{(z-w')^2\,(x-w')^2\,(x'-w)^2\,(y-z)^2}\right.$$
$$\left.+\,\frac{(x-z)\cdot(y-z)\,(x-w')\cdot(x'-w)}{2\,(x-z)^2\,(x-w')^2\,(x'-w)^2\,(y-z)^2}\right]\times\Big[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x)-f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,\rho^{e'}(x)\Big]\times$$
$$\Big[U^{ac}(w)\,\{\rho^c(x'),\rho^b(y)\}-U^{bc}(z)\,U^{ad}(x')\,U^{ce}(y)\,\{\rho^d(x'),\rho^e(y)\}\Big] \tag{2.1}$$

We write it as follows:

$$\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Big[$$
$$-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^b(y)\,\rho^c(x)\,U^{de}(w')\,U^{ad}(w)$$
$$+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{be}(z)\,U^{ad}(w')\,U^{ch}(x)$$
$$+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^l(y)\,\rho^c(x)\,U^{eh}(x')\,U^{bd}(z)\,U^{dl}(y)\,U^{ae}(w)$$
$$-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)$$
$$+\,2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^b(y)\,\rho^c(x)\,U^{de}(w')\,U^{ad}(w)$$
$$-\,2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{eb}(z)\,U^{ch}(x)$$
$$-\,2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^l(y)\,\rho^c(x)\,U^{eh}(x')\,U^{bd}(z)\,U^{dl}(y)\,U^{ea}(w)$$
$$+\,2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{da}(x')\,U^{be}(y)\,U^{ch}(x)$$
$$+\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\,\rho^e(x)\,\rho^b(y)\,U^{ad}(w')\,U^{de}(w)$$
$$-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\,\rho^h(x)\,\rho^l(y)\,U^{ad}(w')\,U^{be}(z)\,U^{dh}(x)\,U^{el}(y)$$
$$-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^h(x)\,\rho^d(y)\,U^{bd}(z)\,U^{ce}(x')\,U^{ah}(w)$$
$$+\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{cd}(x')\,U^{ae}(x)\,U^{bh}(y)$$
$$-\,2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\,\rho^e(x)\,\rho^b(y)\,U^{ad}(w')\,U^{de}(w)$$
$$+\,2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\,\rho^h(x)\,\rho^l(y)\,U^{ad}(w')\,U^{be}(z)\,U^{dh}(x)\,U^{el}(y)$$
$$+\,2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^h(x)\,\rho^d(y)\,U^{ce}(x')\,U^{db}(z)\,U^{ah}(w)$$
$$-\,2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{cd}(x')\,U^{ae}(x)\,U^{bh}(y)\Big] \tag{2.2}$$

2.2 Row II

Only 4 ρ's

2.3 Row III

Only 4 ρ's

2.4 Row IV

Only 2 ρ's

3 Group III (with c.c.)

3.1 Row I

$$\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{2}{(2\pi)^3}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{z,w,w'} e^{-ik(w'-w)}\Big[\bar{\mathbb{A}}_i^{\dagger\,(1)\,a}(k^+,-w')\bar{\mathbb{A}}_j^{\dagger\,(1)\,c'}(p^+,-z)U^{c'b}(z)$$
$$-\,\mathbb{A}_i^{\dagger\,(1)\,c'}(k^+,-w')U^{bd'}(z)\bar{\mathbb{A}}_j^{\dagger\,(1)\,d'}(p^+,-z)U^{c'a}(w')\Big]\times\Big[U^{ac}(w)\mathbb{B}_{2ji}^{bc}(k^+,-w;p^+,-z)-U^{\dagger\,bc}(z)\bar{\mathbb{B}}_{2ji}^{ca}(k^+,-w;p^+,-z)\Big]$$

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\,\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\left[\frac{(x-w)\cdot(x'-w')\,(z-w)\cdot(y-z)}{(z-w)^2\,(x-w)^2\,(x'-w')^2\,(y-z)^2}\right.$$
$$\left.+\,\frac{(x-z)\cdot(y-z)\,(x-w)\cdot(x'-w')}{2\,(x-z)^2\,(x-w)^2\,(x'-w')^2\,(y-z)^2}\right]\times\Big[U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,\rho^{b'}(x')\rho^{d'}(y) - U^{bd'}(z)\,U^{d'e'}(y)U^{ac'}(w')\,\rho^{c'}(x')\rho^{e'}(y)\Big]\times$$
$$\Big[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\Big] \tag{3.1}$$

We can write it as follows:

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\,\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\Big[-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)$$
$$+\,\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\Big]\times\Big[U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,\rho^{b'}(x')\rho^{d'}(y) - U^{bd'}(z)\,U^{d'e'}(y)U^{ac'}(w')\,\rho^{c'}(x')\rho^{e'}$$
$$\Big[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\Big] \tag{3.2}$$

3.2 Row II

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{2}{(2\pi)^3}\int\frac{dp^+}{2\pi}\int_{z,w,w'} e^{-ik(w'-w)}\Big[\bar{\mathbb{A}}_j^{\dagger\,(1)\,c'}(p^+,-z)\bar{\mathbb{A}}_i^{\dagger\,(1)\,a}(k^+,-w')U^{bc'}(z) - \mathbb{A}_j^{\dagger\,(1)\,b}(p^+,-z)\bar{\mathbb{A}}_i^{\dagger\,(1)\,a}(k^+,-w')\Big]\times$$
$$\Big[U^{ac}(w)\mathbb{B}_{2ji}^{bc}(k^+,-w;p^+,-z) - U^{\,bc}(z)\bar{\mathbb{B}}_{2ji}^{ca}(k^+,-w;p^+,-z)\Big]$$

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\,\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\left[\frac{(x-w)\cdot(x'-w')\,(z-w)\cdot(y-z)}{(z-w)^2\,(x-w)^2\,(x'-w')^2\,(y-z)^2}\right.$$
$$\left.+\,\frac{(x-z)\cdot(y-z)\,(x-w)\cdot(x'-w')}{2\,(x-z)^2\,(x-w)^2\,(x'-w')^2\,(y-z)^2}\right]\times\Big[U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,\rho^{d'}(y)\rho^{b'}(x') - U^{ab'}(x')\,\rho^b(y)\rho^{b'}(x')\Big]\times$$
$$\Big[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\Big] \tag{3.3}$$

We write it as follows:

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\,\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\Big[-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)$$
$$+\,\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\Big]\times\Big[U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,\rho^{d'}(y)\rho^{b'}(x') - U^{ab'}(x')\,\rho^b(y)\rho^{b'}(x')\Big]\times$$
$$\Big[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\Big] \tag{3.4}$$

3.3 Row III

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{4}{(2\pi)^3}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{z,w,w'} e^{-ik(w'-w)}\int_{y'}\Big[\mathbb{A}_{k'}^{\dagger\,(1)\,e'}(p^++k^+,-y')U^{\,e'c'}(y')U^{d'b}(z)\bar{\mathbb{C}}_{1jik'}^{\dagger\,d'ac'}(p^++k^+,y'-w';p^+,w'-z)$$
$$-\,\bar{\mathbb{A}}_{k'}^{\dagger\,(1)\,c'}(p^++k^+,-y')\bar{\mathbb{C}}_{1jik'}^{\dagger\,d'ac'}(p^++k^+,y'-w';p^+,w'-z)U^{d'b}(z)\Big]\times\Big[U^{ac}(w)\mathbb{B}_{2ji}^{bc}(k^+,-w;p^+,-z)$$
$$-\,U^{\,bc}(z)\bar{\mathbb{B}}_{2ji}^{ca}(k^+,-w;p^+,-z)\Big]$$

We write it in two parts:

3.3.1 First part:

$$\frac{d^3N_{1\,\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\Big[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\Big[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\Big] \tag{3.5}$$

We write it as follows:

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{16\pi^5}\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{w,x',w',z,y,x} e^{-ik\cdot(w'-w)}\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)$$
$$\Big[U^{e'c'}(w')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\big[U^{ac}(w)\,\{\rho^c(x),\rho^b(y)\} - U^{bc}(z)\,U^{ad}(x)\,U^{ce}(y)\,\{\rho^d(x),\rho^e(y)\}\big] \tag{3.6}$$

3.3.2 Second part:

Only 2 ρ's

3.4 Row IV

Only 4 ρ's

3.5 Row V

$$\frac{d^3N_{\mathrm{NLO}}}{d^3k} = -\frac{2}{(2\pi)^3}\int\frac{dp^+}{2\pi}\int_{z,w,w'} e^{-ik(w'-w)}\int_{y'}\Big[\mathbb{A}^{\dagger\,(1)\,e'}_{k'}(p^+ + k^+,-y')U^{\,e'c'}(y')U^{d'b}(z)\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^+ + k^+,y'-w';p^+,w'-z)$$
$$-\bar{\mathbb{A}}^{\dagger\,(1)\,c'}_{k'}(p^+ + k^+,-y')\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^+ + k^+,y'-w';p^+,w'-z)U^{d'b}(z)\Big]\times\Big[U^{bc}(z)\bar{\mathbb{A}}^{(1)\,c}_{j}(p^+,-z)\bar{\mathbb{A}}^{(1)\,a}_{i}(k^+,-w)$$
$$-U^{ac}(w)\bar{\mathbb{A}}^{(1)\,d}_{j}(p^+,-z)U^{db}(z)\mathbb{A}^{(1)\,c}_{i}(k^+,-w)\Big]$$

$$\frac{d^3N_{\mathrm{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'ca}}{8\pi^5}\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{w,w',z,y,x,x'} e^{-ik\cdot(w'-w)}\,\frac{(w'-z)\cdot(y-z)\,(x'-w')\cdot(x-w)}{(w'-z)^2\,(x'-w')^2\,(x-w)^2\,(y-z)^2}$$
$$\Big[U^{e'c'}(w')\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\big[U^{ce}(y)\,U^{ad}(x)\,\rho^e(y)\rho^d(x) - U^{ad}(w)\,U^{ce}(y)\,\rho^e(y)\rho^d(x)\big] \tag{3.7}$$

We can write it as follows:

$$\frac{d^3N_{\mathrm{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'ca}}{8\pi^5}\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{w,w',z,y,x,x'} e^{-ik\cdot(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)$$
$$\Big[U^{e'c'}(w')\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\big[U^{ce}(y)\,U^{ad}(x)\,\rho^e(y)\rho^d(x) - U^{ad}(w)\,U^{ce}(y)\,\rho^e(y)\rho^d(x)\big] \tag{3.8}$$

3.6 Row VI

$$\frac{d^3N_{\mathrm{NLO}}}{d^3k} = -\frac{2}{(2\pi)^3}\int\frac{dp^+}{2\pi}\int_{z,w,w'} e^{-ik(w'-w)}\int_{y'}\Big[\mathbb{A}^{\dagger\,(1)\,e'}_{k'}(p^+ + k^+,-y')U^{\,e'c'}(y')U^{d'b}(z)\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^+ + k^+,y'-w';p^+,w'-z)$$
$$-\bar{\mathbb{A}}^{\dagger\,(1)\,c'}_{k'}(p^+ + k^+,-y')\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^+ + k^+,y'-w';p^+,w'-z)U^{d'b}(z)\Big]\times\Big[U^{bc}(z)\bar{\mathbb{A}}^{(1)\,a}_{i}(k^+,-w)\bar{\mathbb{A}}^{(1)\,c}_{j}(p^+,-z)$$
$$-\bar{\mathbb{A}}^{(1)\,a}_{i}(k^+,-w)\mathbb{A}^{(1)\,b}_{j}(p^+,-z)\Big]$$

$$\frac{d^3N_{\mathrm{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{8\pi^5}\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{w,w',z,y,x,x'} e^{-ik\cdot(w'-w)}\,\frac{(w'-z)\cdot(y-z)\,(x'-w')\cdot(x-w)}{(w'-z)^2\,(x'-w')^2\,(x-w)^2\,(y-z)^2}$$
$$\Big[U^{e'c'}(w')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\big[U^{bc}(z)\,U^{ad}(x)\,U^{ce}(y)\,\rho^d(x)\,\rho^e(y) - U^{ad}(x)\,\rho^d(x)\,\rho^b(y)\big] \tag{3.9}$$

We can write it as follows:

$$\frac{d^3N_{\mathrm{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{8\pi^5}\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{w,w',z,y,x,x'} e^{-ik\cdot(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)$$
$$\Big[U^{e'c'}(w')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\big[U^{bc}(z)\,U^{ad}(x)\,U^{ce}(y)\,\rho^d(x)\,\rho^e(y) - U^{ad}(x)\,\rho^d(x)\,\rho^b(y)\big] \tag{3.10}$$

4 All three ρ term with c.c.

4.1 From four ρ's

$$\frac{d^3N_{a\,\mathrm{NLO}}}{d^3k} = -\frac{i}{(2\pi)^3}\frac{g^4}{8\pi^3}\frac{1}{2\pi^2}\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Big[$$
$$\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^a(y)\rho^c(x)\,U^{ed}(w')U^{eb}(w)\,\alpha$$
$$-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^e(x')\rho^a(y)\rho^c(x)\,U^{db}(w)U^{de}(x')\,\alpha$$
$$+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^c(x')\rho^a(y)\rho^e(x)\,U^{db}(w')U^{de}(x)\,\alpha'$$
$$-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^c(x')\rho^a(y)\rho^e(x)U^{db}(w')U^{de}(w)\,\alpha'\Big] \tag{4.1}$$

$$\frac{d^3N_{b\,\mathrm{NLO}}}{d^3k} = \frac{i}{(2\pi)^3}\frac{g^4}{8\pi^3}\frac{1}{2\pi^2}\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Big[$$
$$-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^h(y)\rho^e(x)U^{bd}(w')U^{ce}(x)U^{ah}(y)\,\alpha$$
$$+\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^e(x)\rho^h(y)U^{cd}(x')U^{be}(w)U^{ah}(y)\,\alpha'$$
$$-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^e(x)\rho^h(y)U^{cd}(x')U^{be}(x)U^{ah}(y)\,\alpha'$$
$$+\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^e(x')\rho^h(y)\rho^d(x)U^{cd}(x)U^{be}(x')U^{ah}(y)\,\alpha\Big] \tag{4.2}$$

4.2 Genuine terms

4.2.1 Group I, Row II

We write it in two parts:

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{ig^4}{16\pi^5}\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Big[$$
$$-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(x')\,U^{be}(x)\,U^{ch}(y)\,\alpha$$
$$+\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\,\alpha$$
$$+\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(w')\,U^{be}(x)\,U^{ch}(y)\,\alpha$$
$$-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{be}(y)\,U^{ch}(x)\,\alpha$$
$$-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^b(x)\,\rho^c(y)\,U^{de}(x')\,U^{da}(w)\,\beta$$
$$+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^b(y)\,\rho^c(x)\,U^{de}(x')\,U^{da}(w)\,\beta$$
$$+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(x')\,U^{be}(x)\,U^{ch}(y)\,\beta$$
$$-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\,\beta$$
$$+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^b(x)\,\rho^c(y)\,U^{de}(w')\,U^{da}(w)\,\beta$$
$$-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^b(y)\,\rho^c(x)\,U^{de}(w')\,U^{da}(w)\,\beta$$
$$-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(w')\,U^{be}(x)\,U^{ch}(y)\,\beta$$
$$+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{be}(y)\,U^{ch}(x)\,\beta\Big] \tag{4.3}$$

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Bigg[\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(x)\,U^{be}(x')\,U^{ch}(y)\,\alpha' \\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')\,\alpha' \\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(w)\,U^{be}(x')\,U^{ch}(y)\,\alpha' \\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(w)\,U^{be}(y)\,U^{ch}(x')\,\alpha' \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(y)\,\rho^b(x')\,\rho^e(x)\,U^{de}(x)\,U^{da}(w')\,\beta' \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\,\rho^b(y)\,\rho^e(x)\,U^{de}(x)\,U^{da}(w')\,\beta' \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(x)\,U^{be}(x')\,U^{ch}(y)\,\beta' \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')\,\beta' \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(y)\,\rho^b(x')\,\rho^e(x)\,U^{de}(w)\,U^{da}(w')\,\beta' \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\,\rho^b(y)\,\rho^e(x)\,U^{de}(w)\,U^{da}(w')\,\beta' \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(w)\,U^{be}(x')\,U^{ch}(y)\,\beta' \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(w)\,U^{be}(y)\,U^{ch}(x')\,\beta' \qquad (4.4)
\end{aligned}
$$

4.2.2 Group I, Row III

$$
\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Bigg[\\
&\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^e(x)\rho^h(y)\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(z)\,\alpha \\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^e(x)\rho^h(y)\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(y)\,\alpha \\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^e(x)\rho^h(y)\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(z)\,\alpha \\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^e(x)\rho^h(y)\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(y)\,\alpha \\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(z)\,\alpha' \\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(y)\,\alpha' \\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(z)\,\alpha' \\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(y)\,\alpha' \\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{ce}(y)\,U^{bh}(w)\,\beta \\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{ce}(y)\,U^{bh}(x)\,\beta \\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{ce}(y)\,U^{bh}(w)\,\beta \\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{ce}(y)\,U^{bh}(x)\,\beta \\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{ce}(y)\,U^{bh}(w')\,\beta' \\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{ce}(y)\,U^{bh}(x')\,\beta' \\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(y)\,U^{bh}(w')\,\beta' \\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(y)\,U^{bh}(x')\,\beta'\Bigg] \qquad (4.5)
\end{aligned}
$$

4.2.3 Group I, Row IV

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Bigg[\\
&\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(x')\,U^{da}(w)\,U^{hb}(z)\,U^{hl}(y)\,\alpha \\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\,\alpha \\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(w')\,U^{da}(w)\,U^{hb}(z)\,U^{hl}(y)\,\alpha \\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{be}(y)\,U^{ch}(x)\,\alpha \\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(x)\,U^{da}(w')\,U^{hb}(z)\,U^{hl}(y)\,\alpha' \\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{ch}(x')\,U^{be}(y)\,\alpha' \\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(w)\,U^{da}(w')\,U^{hb}(z)\,U^{hl}(y)\,\alpha' \\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(w)\,U^{ch}(x')\,U^{be}(y)\,\alpha' \\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(x')\,U^{da}(w)\,U^{hb}(z)\,U^{hl}(y)\,\beta \\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(w')\,U^{da}(w)\,U^{hb}(z)\,U^{hl}(y)\,\beta \\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\,\beta \\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{be}(y)\,U^{ch}(x)\,\beta \\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(x)\,U^{da}(w')\,U^{hb}(z)\,U^{hl}(y)\,\beta' \\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(w)\,U^{da}(w')\,U^{hb}(z)\,U^{hl}(y)\,\beta' \\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{be}(y)\,U^{ad}(w)\,U^{ch}(x')\,\beta' \\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{be}(y)\,U^{ad}(x)\,U^{ch}(x')\,\beta'\Bigg]
\end{aligned}
\tag{4.6}
$$

4.2.4 Group I, Row V

$$
\begin{aligned}
\frac{d^3N_{1\,\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Bigg[\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\big\{\rho^h(x),\rho^e(y)\big\}\,U^{ad}(x')\,U^{be}(z)\,U^{ch}(w)\,\beta \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\big\{\rho^h(x),\rho^e(y)\big\}\,U^{ad}(w')\,U^{be}(z)\,U^{ch}(w)\,\beta \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(y)\,\beta \\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(y)\,\beta \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\big\{\rho^h(x'),\rho^e(y)\big\}\rho^d(x)\,U^{ad}(x)\,U^{be}(z)\,U^{ch}(w')\,\beta' \\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\big\{\rho^h(x'),\rho^e(y)\big\}\rho^d(x)\,U^{ad}(w)\,U^{be}(z)\,U^{ch}(w')\,\beta' \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\big\{\rho^e(x'),\rho^h(y)\big\}\rho^d(x)\,U^{ad}(w)\,U^{bh}(y)\,U^{ce}(x')\,\beta' \\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\big\{\rho^e(x'),\rho^h(y)\big\}\rho^d(x)\,U^{ad}(x)\,U^{bh}(y)\,U^{ce}(x')\,\beta'\Bigg]
\end{aligned}
\tag{4.7}
$$

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Bigg[\\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{ad}(x')\,U^{be}(w)\,U^{ch}(z)\,\beta \\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\big\{\rho^h(x),\rho^e(y)\big\}\,U^{ad}(x')\,U^{ce}(y)\,U^{bh}(x)\,\beta \\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{ad}(w')\,U^{be}(w)\,U^{ch}(z)\,\beta \\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\big\{\rho^h(x),\rho^e(y)\big\}\,U^{ad}(w')\,U^{ce}(y)\,U^{bh}(x)\,\beta \\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\big\{\rho^e(x'),\rho^h(y)\big\}\rho^d(x)\,U^{ad}(x)\,U^{be}(w')\,U^{ch}(z)\,\beta' \\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\big\{\rho^h(x'),\rho^e(y)\big\}\rho^d(x)\,U^{ad}(x)\,U^{ce}(y)\,U^{bh}(x')\,\beta' \\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\big\{\rho^e(x'),\rho^h(y)\big\}\rho^d(x)\,U^{ad}(w)\,U^{ch}(z)\,U^{be}(w')\,\beta' \\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\big\{\rho^h(x'),\rho^e(y)\big\}\rho^d(x)\,U^{ad}(w)\,U^{bh}(x')\,U^{ce}(y)\,\beta'\Bigg]
\end{aligned}
\tag{4.8}
$$

4.2.5 Group II, Row I

$$
\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Bigg[\\
-\frac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\big\{\rho^e(x'),\rho^b(y)\big\}\rho^c(x)\,U^{de}(w')\,U^{da}(w)\,\alpha \\
+\frac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\big\{\rho^d(x'),\rho^e(y)\big\}\rho^h(x)\,U^{be}(z)\,U^{ad}(w')\,U^{ch}(x)\,\alpha \\
+\frac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\big\{\rho^h(x'),\rho^l(y)\big\}\rho^c(x)\,U^{eh}(x')\,U^{db}(z)\,U^{dl}(y)\,U^{ea}(w)\,\alpha \\
-\frac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\big\{\rho^d(x'),\rho^e(y)\big\}\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\,\alpha \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\big\{\rho^e(x'),\rho^b(y)\big\}\rho^c(x)\,U^{de}(w')\,U^{da}(w)\,\beta \\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\big\{\rho^d(x'),\rho^e(y)\big\}\rho^h(x)\,U^{ad}(w')\,U^{be}(z)\,U^{ch}(x)\,\beta \\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\big\{\rho^h(x'),\rho^l(y)\big\}\rho^c(x)\,U^{eh}(x')\,U^{db}(z)\,U^{dl}(y)\,U^{ea}(w)\,\beta \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\big\{\rho^d(x'),\rho^e(y)\big\}\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\,\beta \\
+\frac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\big\{\rho^e(x),\rho^b(y)\big\}\,U^{da}(w')\,U^{de}(w)\,\alpha' \\
-\frac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\big\{\rho^h(x),\rho^l(y)\big\}\,U^{da}(w')\,U^{eb}(z)\,U^{dh}(x)\,U^{el}(y)\,\alpha' \\
-\frac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\big\{\rho^h(x),\rho^d(y)\big\}\,U^{bd}(z)\,U^{ce}(x')\,U^{ah}(w)\,\alpha' \\
+\frac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{cd}(x')\,U^{ae}(x)\,U^{bh}(y)\,\alpha' \\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\big\{\rho^e(x),\rho^b(y)\big\}\,U^{da}(w')\,U^{de}(w)\,\beta' \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\big\{\rho^h(x),\rho^l(y)\big\}\,U^{da}(w')\,U^{eb}(z)\,U^{dh}(x)\,U^{el}(y)\,\beta' \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\big\{\rho^h(x),\rho^d(y)\big\}\,U^{ce}(x')\,U^{bd}(z)\,U^{ah}(w)\,\beta' \\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{cd}(x')\,U^{ae}(x)\,U^{bh}(y)\,\beta'\Bigg]
\end{aligned}
\tag{4.9}
$$

4.2.6 Group III, Row I

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Bigg[\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(x')\,U^{hl}(y)\,U^{hb}(z)\,U^{da}(w)\,\alpha \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\,\alpha \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^l(x')\,\rho^e(y)\,\rho^c(x)\,U^{db}(z)\,U^{de}(y)\,U^{hl}(w')\,U^{ha}(w)\,\alpha \\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^d(y)\,\rho^h(x)\,U^{ae}(w')\,U^{bd}(y)\,U^{ch}(x)\,\alpha \\
&+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(x')\,U^{hl}(y)\,U^{hb}(z)\,U^{da}(w)\,\beta \\
&- 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\,\beta \\
&- 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^l(x')\,\rho^e(y)\,\rho^c(x)\,U^{db}(z)\,U^{de}(y)\,U^{hl}(w')\,U^{ha}(w)\,\beta \\
&+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^d(y)\,\rho^h(x)\,U^{ae}(w')\,U^{bd}(y)\,U^{ch}(x)\,\beta \\
&+ \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(x)\,U^{hl}(y)\,U^{hb}(z)\,U^{da}(w')\alpha' \\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')\alpha' \\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\,\rho^e(y)\,\rho^l(x)\,U^{db}(z)\,U^{de}(y)\,U^{hl}(w)\,U^{ha}(w')\alpha' \\
&+ \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^d(y)\,\rho^e(x)\,U^{bd}(y)\,U^{ae}(w)\,U^{ch}(x')\alpha' \\
&- 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(x)\,U^{hl}(y)\,U^{hb}(z)\,U^{da}(w')\beta' \\
&+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')\beta' \\
&+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\,\rho^e(y)\,\rho^l(x)\,U^{de}(y)\,U^{hl}(w)\,U^{db}(z)\,U^{ha}(w')\beta' \\
&- 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^d(y)\,\rho^e(x)\,U^{bd}(y)\,U^{ae}(w)\,U^{ch}(x')\beta'\Bigg]
\end{aligned}
\tag{4.10}
$$

4.2.7 Group III, Row II

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Bigg[\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(y)\,\rho^l(x')\,\rho^c(x)\,U^{de}(y)\,U^{hl}(x')\,U^{db}(z)\,U^{ha}(w)\,\alpha \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(y)\,\rho^e(x')\,\rho^h(x)\,U^{bd}(y)\,U^{ae}(x')\,U^{ch}(x)\,\alpha \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^b(y)\,\rho^e(x')\,\rho^c(x)\,U^{de}(x')\,U^{da}(w)\,\alpha \\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(y)\,\rho^d(x')\,\rho^h(x)\,U^{ad}(x')\,U^{be}(z)\,U^{ch}(x)\,\alpha \\
&+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(y)\,\rho^l(x')\,\rho^c(x)\,U^{de}(y)\,U^{hl}(x')\,U^{db}(z)\,U^{ha}(w)\,\beta \\
&- 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(y)\,\rho^e(x')\,\rho^h(x)\,U^{bd}(y)\,U^{ae}(x')\,U^{ch}(x)\,\beta \\
&- 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^b(y)\,\rho^e(x')\,\rho^c(x)\,U^{de}(x')\,U^{da}(w)\,\beta \\
&+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(y)\,\rho^d(x')\,\rho^h(x)\,U^{ad}(x')\,U^{be}(z)\,U^{ch}(x)\,\beta \\
&+ \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\,\rho^l(x)\,\rho^e(y)\,U^{de}(y)\,U^{hl}(x)\,U^{db}(z)\,U^{ha}(w')\,\alpha' \\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^e(x)\,\rho^d(y)\,U^{bd}(y)\,U^{ae}(x)\,U^{ch}(x')\,\alpha' \\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\,\rho^e(x)\,\rho^b(y)\,U^{de}(x)\,U^{da}(w')\,\alpha' \\
&+ \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^d(x)\,\rho^e(y)\,U^{ad}(x)\,U^{be}(z)\,U^{ch}(x')\,\alpha' \\
&- 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\,\rho^l(x)\,\rho^e(y)\,U^{de}(y)\,U^{hl}(x)\,U^{db}(z)\,U^{ha}(w')\,\beta' \\
&+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^e(x)\,\rho^d(y)\,U^{bd}(y)\,U^{ae}(x)\,U^{ch}(x')\,\beta' \\
&+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^c(x')\,\rho^e(x)\,\rho^b(y)\,U^{de}(x)\,U^{da}(w')\,\beta' \\
&- 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^d(x)\,\rho^e(y)\,U^{ad}(x)\,U^{be}(z)\,U^{ch}(x')\,\beta'\Bigg]
\end{aligned}
\tag{4.11}
$$

4.2.8 Group III, Row III

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{ig^4}{16\pi^5}\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Bigg[$$
$$+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\left\{\rho^h(x'),\rho^e(y)\right\}\rho^d(x)\,U^{ad}(w)\,U^{be}(z)\,U^{ch}(w')\,\beta$$
$$-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\left\{\rho^e(x'),\rho^h(y)\right\}\rho^d(x)\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(y)\,\beta$$
$$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\left\{\rho^h(x'),\rho^d(y)\right\}\rho^e(x)\,U^{bd}(z)\,U^{ae}(x)\,U^{ch}(w')\,\beta$$
$$+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\left\{\rho^e(x'),\rho^h(y)\right\}\rho^d(x)\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(y)\,\beta$$
$$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\left\{\rho^h(x),\rho^e(y)\right\}U^{ad}(w')\,U^{be}(z)\,U^{ch}(w)\,\beta'$$
$$+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\left\{\rho^e(x),\rho^h(y)\right\}U^{ad}(w')\,U^{ce}(x)\,U^{bh}(y)\,\beta'$$
$$+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\left\{\rho^h(x),\rho^d(y)\right\}U^{bd}(z)\,U^{ae}(x')\,U^{ch}(w)\,\beta'$$
$$-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\left\{\rho^e(x),\rho^h(y)\right\}U^{ad}(x')\,U^{ce}(x)\,U^{bh}(y)\,\beta'\Bigg] \tag{4.12}$$

4.2.9 Group III, Row V

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{ig^4}{16\pi^5}\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Bigg[$$
$$2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(w)\,U^{be}(y)\,U^{ch}(x')\,\beta$$
$$-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^h(y)\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(w')\,U^{bh}(y)\,\beta$$
$$-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{be}(y)\,U^{ad}(x)\,U^{ch}(x')\,\beta$$
$$+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^h(y)\,\rho^d(x)\,U^{ad}(x)\,U^{ce}(w')\,U^{bh}(y)\,\beta$$
$$-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{be}(y)\,U^{ch}(x)\,\beta'$$
$$+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^h(y)\,\rho^e(x)\,U^{ad}(w')\,U^{ce}(w)\,U^{bh}(y)\,\beta'$$
$$+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{be}(y)\,U^{ad}(x')\,U^{ch}(x)\,\beta'$$
$$-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^h(y)\,\rho^e(x)\,U^{ad}(x')\,U^{ce}(w)\,U^{bh}(y)\,\beta'\Bigg] \tag{4.13}$$

4.2.10 Group III, Row VI

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{ig^4}{16\pi^5}\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Bigg[$$
$$+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(y)\,\beta$$
$$-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(y)\,\rho^h(x')\,\rho^d(x)\,U^{ad}(w)\,U^{be}(z)\,U^{ch}(x')\,\beta$$
$$-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{bh}(y)\,U^{ad}(x)\,U^{ce}(x')\,\beta$$
$$+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(y)\,\rho^h(x')\,\rho^e(x)\,U^{ae}(x)\,U^{ch}(x')\,U^{bd}(z)\,\beta$$
$$-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(w')\,U^{bh}(y)\,U^{ce}(x)\,\beta'$$
$$+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^h(x)\,\rho^e(y)\,U^{ad}(w')\,U^{be}(z)\,U^{ch}(x)\,\beta'$$
$$+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{bh}(y)\,U^{ad}(x')\,U^{ce}(x)\,\beta'$$
$$-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\,\rho^e(x')\,\rho^h(x)\,\rho^d(y)\,U^{ae}(x')\,U^{ch}(x)\,U^{bd}(z)\,\beta'\Bigg] \tag{4.14}$$

5 Cancellation of all three ρ's

Verification (added). The complete three-ρ $\log(k^+/\Lambda)$ set collected in the previous section ("From four ρ's" + all "Genuine terms", exactly as written there) has been verified to cancel *identically as an integrand* – after symmetrization $\rho\rho = \frac{1}{2}\{\rho,\rho\}+\frac{1}{2}[\rho,\rho]$ with $[\rho^a(x),\rho^b(y)] = if^{abc}\delta^{(2)}(x-y)\rho^c(x)$, every kernel $\otimes$ colour structure sums to zero for *arbitrary* N_c (checked symbolically, and numerically with exact SU(2) and SU(3) adjoint Wilson lines). The β and β' sub-sums below were left incomplete in the draft; they are covered by this global check, which does not rely on the sector-by-sector displays of this section.

5.1 α terms

5.1.1 Two Wilson lines

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{i\,f^{abc}}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)$$
$$\times\Big[-\rho^d(x')\rho^a(y)\rho^c(x)\,U^{ed}(w')U^{eb}(w)-\frac{1}{2}\left\{\rho^e(x'),\rho^b(y)\right\}\rho^c(x)\,U^{de}(w')\,U^{da}(w)$$
$$+\rho^e(x')\rho^a(y)\rho^c(x)\,U^{db}(w)U^{de}(x')+\rho^b(y)\,\rho^e(x')\,\rho^c(x)\,U^{de}(x')\,U^{da}(w)\Big]=0 \tag{5.1}$$

5.1.2 Three Wilson lines

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{i\,f^{abc}}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)$$
$$\times\Big[-\rho^d(x')\rho^h(y)\rho^e(x)U^{bd}(w')U^{ce}(x)U^{ah}(y)+\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(w')\,U^{be}(x)\,U^{ch}(y)$$
$$-\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{be}(y)\,U^{ch}(x)+\rho^d(x')\rho^e(x)\rho^h(y)\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(y)$$
$$+\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{be}(y)\,U^{ch}(x)-\rho^e(x')\,\rho^d(y)\,\rho^h(x)\,U^{ae}(w')\,U^{bd}(y)\,U^{ch}(x)$$
$$+\rho^e(x')\rho^h(y)\rho^d(x)U^{cd}(x)U^{be}(x')U^{ah}(y)-\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(x')\,U^{be}(x)\,U^{ch}(y)$$
$$+\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)-\rho^d(x')\rho^e(x)\rho^h(y)\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(y)$$
$$-\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)-\frac{1}{2}\left\{\rho^d(x'),\rho^e(y)\right\}\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)$$
$$+\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)+\rho^d(y)\,\rho^e(x')\,\rho^h(x)\,U^{bd}(y)\,U^{ae}(x')\,U^{ch}(x)$$
$$+\rho^d(x')\rho^e(x)\rho^h(y)\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(z)-\rho^e(y)\,\rho^d(x')\,\rho^h(x)\,U^{ad}(x')\,U^{be}(z)\,U^{ch}(x)$$
$$-\rho^d(x')\rho^e(x)\rho^h(y)\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(z)+\frac{1}{2}\left\{\rho^d(x'),\rho^e(y)\right\}\rho^h(x)\,U^{be}(z)\,U^{ad}(w')\,U^{ch}(x)\Big]=0 \tag{5.2}$$

5.1.3 Four Wilson lines

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{i\,f^{abc}}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)$$
$$\times\Big[+\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(x')\,U^{da}(w)\,U^{hb}(z)\,U^{hl}(y)+\frac{1}{2}\left\{\rho^h(x'),\rho^l(y)\right\}\rho^c(x)\,U^{eh}(x')\,U^{db}(z)\,U^{dl}(y)\,U^{ea}(w)$$
$$-\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(x')\,U^{hl}(y)\,U^{hb}(z)\,U^{da}(w)-\rho^e(y)\,\rho^l(x')\,\rho^c(x)\,U^{de}(y)\,U^{hl}(x')\,U^{db}(z)\,U^{ha}(w)$$
$$-\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(w')\,U^{da}(w)\,U^{hb}(z)\,U^{hl}(y)+\rho^l(x')\,\rho^e(y)\,\rho^c(x)\,U^{db}(z)\,U^{de}(y)\,U^{hl}(w')\,U^{ha}(w)\Big]=0 \tag{5.3}$$

5.2 α' terms

5.2.1 Two Wilson lines

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{i\,f^{abc}}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)$$
$$\times\Big[-\rho^c(x')\rho^a(y)\rho^e(x)\,U^{db}(w')U^{de}(x)-\rho^c(x')\,\rho^e(x)\,\rho^b(y)\,U^{de}(x)\,U^{da}(w')$$
$$+\rho^c(x')\rho^a(y)\rho^e(x)U^{db}(w')U^{de}(w)+\frac{1}{2}\,\rho^c(x')\left\{\rho^e(x),\rho^b(y)\right\}U^{da}(w')\,U^{de}(w)\Big]=0 \tag{5.4}$$

5.2.2 Three Wilson lines

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} =& \frac{i\,f^{abc}}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\\
&\times\Big[\rho^d(x')\rho^e(x)\rho^h(y)U^{cd}(x')U^{be}(w)U^{ah}(y)-\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(w)\,U^{be}(x')\,U^{ch}(y)\\
&+\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(w)\,U^{be}(y)\,U^{ch}(x')-\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(y)\\
&-\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(w)\,U^{ch}(x')\,U^{be}(y)+\rho^h(x')\,\rho^d(y)\,\rho^e(x)\,U^{bd}(y)\,U^{ae}(w)\,U^{ch}(x')\\
&-\rho^d(x')\rho^e(x)\rho^h(y)U^{cd}(x')U^{be}(x)U^{ah}(y)+\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(x)\,U^{be}(x')\,U^{ch}(y)\\
&-\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')+\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(y)\\
&+\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{ch}(x')\,U^{be}(y)+\frac{1}{2}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{cd}(x')\,U^{ae}(x)\,U^{bh}(y)\\
&-\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')-\rho^h(x')\,\rho^e(x)\,\rho^d(y)\,U^{bd}(y)\,U^{ae}(x)\,U^{ch}(x')\\
&-\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(z)+\rho^h(x')\,\rho^d(x)\,\rho^e(y)\,U^{ad}(x)\,U^{be}(z)\,U^{ch}(x')\\
&+\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(z)-\frac{1}{2}\,\rho^e(x')\big\{\rho^h(x),\rho^d(y)\big\}\,U^{bd}(z)\,U^{ce}(x')\,U^{ah}(w)\Big]=0\end{aligned}\tag{5.5}$$

5.2.3 Four Wilson lines

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} =& \frac{i\,f^{abc}}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\\
&\times\Big[-\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(x)\,U^{da}(w')\,U^{hb}(z)\,U^{hl}(y)-\frac{1}{2}\,\rho^c(x')\big\{\rho^h(x),\rho^l(y)\big\}\,U^{da}(w')\,U^{eb}(z)\,U^{dh}(x)\,U^{el}(y)\\
&+\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(x)\,U^{hl}(y)\,U^{hb}(z)\,U^{da}(w')+\rho^c(x')\,\rho^l(x)\,\rho^e(y)\,U^{de}(y)\,U^{hl}(x)\,U^{db}(z)\,U^{ha}(w')\\
&+\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(w)\,U^{da}(w')\,U^{hb}(z)\,U^{hl}(y)-\rho^c(x')\,\rho^e(y)\,\rho^l(x)\,U^{db}(z)\,U^{de}(y)\,U^{hl}(w)\,U^{ha}(w')\Big]=0\end{aligned}\tag{5.6}$$

5.3 β terms

5.3.1 Two Wilson lines

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} =& \frac{i\,f^{abc}}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\\
&\times\Big[-\rho^e(x')\,\rho^b(x)\,\rho^c(y)\,U^{de}(x')\,U^{da}(w)+\rho^e(x')\,\rho^b(y)\,\rho^c(x)\,U^{de}(x')\,U^{da}(w)-2\,\rho^b(y)\,\rho^e(x')\,\rho^c(x)\,U^{de}(x')\,U^{da}(w)\\
&+\rho^e(x')\,\rho^b(x)\,\rho^c(y)\,U^{de}(w')\,U^{da}(w)-\rho^e(x')\,\rho^b(y)\,\rho^c(x)\,U^{de}(w')\,U^{da}(w)+\big\{\rho^e(x'),\rho^b(y)\big\}\rho^c(x)\,U^{de}(w')\,U^{da}(w)\Big]=0\end{aligned}\tag{5.7}$$

5.3.2 Three Wilson lines

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} =& \frac{i\,f^{abc}}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\\
&\times\Big[+\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(x')\,U^{be}(x)\,U^{ch}(y)-\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\\
&+2\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{ce}(y)\,U^{bh}(x)+2\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\\
&+\frac{1}{2}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(y)-\frac{1}{2}\,\rho^d(x')\big\{\rho^h(x),\rho^e(y)\big\}\,U^{ad}(x')\,U^{ce}(y)\,U^{bh}(x)\\
&+\big\{\rho^d(x'),\rho^e(y)\big\}\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)-2\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\\
&-2\,\rho^d(y)\,\rho^e(x')\,\rho^h(x)\,U^{bd}(y)\,U^{ae}(x')\,U^{ch}(x)+\big\{\rho^e(x'),\rho^h(y)\big\}\rho^d(x)\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(y)\\
&-2\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{be}(y)\,U^{ad}(x)\,U^{ch}(x')-2\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{bh}(y)\,U^{ad}(x)\,U^{ce}(x')\\
&-\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(w')\,U^{be}(x)\,U^{ch}(y)+\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{be}(y)\,U^{ch}(x)\\
&-2\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{ce}(y)\,U^{bh}(x)-2\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{be}(y)\,U^{ch}(x)\\
&-\frac{1}{2}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(y)+\frac{1}{2}\,\rho^d(x')\big\{\rho^h(x),\rho^e(y)\big\}\,U^{ad}(w')\,U^{ce}(y)\,U^{bh}(x)\\
&+2\,\rho^e(x')\,\rho^d(y)\,\rho^h(x)\,U^{ae}(w')\,U^{bd}(y)\,U^{ch}(x)+2\,\rho^e(x')\,\rho^h(y)\,\rho^d(x)\,U^{ad}(x)\,U^{ce}(w')\,U^{bh}(y)\\
&-2\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{ce}(y)\,U^{bh}(w)-\big\{\rho^e(x'),\rho^h(y)\big\}\rho^d(x)\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(y)\\
&+2\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(w)\,U^{be}(y)\,U^{ch}(x')+2\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(y)\\
&+2\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{ce}(y)\,U^{bh}(w)-2\,\rho^e(x')\,\rho^h(y)\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(w')\,U^{bh}(y)\\
&-\frac{1}{2}\,\rho^d(x')\big\{\rho^h(x),\rho^e(y)\big\}\,U^{ad}(x')\,U^{be}(z)\,U^{ch}(w)+\frac{1}{2}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{ad}(x')\,U^{be}(w)\,U^{ch}(z)\\
&-2\,\rho^e(y)\,\rho^h(x')\,\rho^d(x)\,U^{ad}(w)\,U^{be}(z)\,U^{ch}(x')+\frac{1}{2}\,\rho^d(x')\big\{\rho^h(x),\rho^e(y)\big\}\,U^{ad}(w')\,U^{be}(z)\,U^{ch}(w)\\
&-\frac{1}{2}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{ad}(w')\,U^{be}(w)\,U^{ch}(z)+\big\{\rho^h(x'),\rho^e(y)\big\}\rho^d(x)\,U^{ad}(w)\,U^{be}(z)\,U^{ch}(w')\\
&-\big\{\rho^d(x'),\rho^e(y)\big\}\rho^h(x)\,U^{ad}(w')\,U^{be}(z)\,U^{ch}(x)-\big\{\rho^h(x'),\rho^d(y)\big\}\rho^e(x)\,U^{bd}(z)\,U^{ae}(x)\,U^{ch}(w')\\
&+2\,\rho^e(y)\,\rho^d(x')\,\rho^h(x)\,U^{ad}(x')\,U^{be}(z)\,U^{ch}(x)+2\,\rho^d(y)\,\rho^h(x')\,\rho^e(x)\,U^{ae}(x)\,U^{ch}(x')\,U^{bd}(z)\Big]=0
\end{aligned}
\tag{5.8}
$$

5.3.3 Four Wilson lines

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} =& \frac{i\,f^{abc}}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\\
&\times\Big[-2\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(x')\,U^{da}(w)\,U^{hb}(z)\,U^{hl}(y)-\big\{\rho^h(x'),\rho^l(y)\big\}\rho^c(x)\,U^{eh}(x')\,U^{db}(z)\,U^{dl}(y)\,U^{ea}(w)\\
&+2\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(x')\,U^{hl}(y)\,U^{hb}(z)\,U^{da}(w)+2\,\rho^e(y)\,\rho^l(x')\,\rho^c(x)\,U^{de}(y)\,U^{hl}(x')\,U^{db}(z)\,U^{ha}(w)\\
&+2\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(w')\,U^{da}(w)\,U^{hb}(z)\,U^{hl}(y)-2\,\rho^l(x')\,\rho^e(y)\,\rho^c(x)\,U^{db}(z)\,U^{de}(y)\,U^{hl}(w')\,U^{ha}(w)\Big]=0
\end{aligned}
\tag{5.9}
$$

5.4 β' terms

5.4.1 Two Wilson lines

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} =& \frac{i\,f^{abc}}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\\
&\times\Big[+\rho^c(y)\,\rho^b(x')\,\rho^e(x)\,U^{de}(x)\,U^{da}(w')-\rho^c(x')\,\rho^b(y)\,\rho^e(x)\,U^{de}(x)\,U^{da}(w')\\
&+2\,\rho^c(x')\,\rho^e(x)\,\rho^b(y)\,U^{de}(x)\,U^{da}(w')-\rho^c(y)\,\rho^b(x')\,\rho^e(x)\,U^{de}(w)\,U^{da}(w')+\rho^c(x')\,\rho^b(y)\,\rho^e(x)\,U^{de}(w)\,U^{da}(w')\\
&-\rho^c(x')\big\{\rho^e(x),\rho^b(y)\big\}\,U^{da}(w')\,U^{de}(w)\Big]=0
\end{aligned}
\tag{5.10}
$$

5.4.2 Four Wilson lines

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} =& \frac{i\,f^{abc}}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\\
&\times\Big[+2\,\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(x)\,U^{da}(w')\,U^{hb}(z)\,U^{hl}(y)+\rho^c(x')\big\{\rho^h(x),\rho^l(y)\big\}\,U^{da}(w')\,U^{eb}(z)\,U^{dh}(x)\,U^{el}(y)\\
&-2\,\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(x)\,U^{hl}(y)\,U^{hb}(z)\,U^{da}(w')-2\,\rho^c(x')\,\rho^l(x)\,\rho^e(y)\,U^{de}(y)\,U^{hl}(x)\,U^{db}(z)\,U^{ha}(w')\\
&-2\,\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(w)\,U^{da}(w')\,U^{hb}(z)\,U^{hl}(y)+2\,\rho^c(x')\,\rho^e(y)\,\rho^l(x)\,U^{de}(y)\,U^{hl}(w)\,U^{db}(z)\,U^{ha}(w')\Big]=0\end{aligned}\tag{5.11}$$

5.4.3 Three Wilson lines

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} =& \frac{i\,f^{abc}}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\\
&\times\Big[-\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(x)\,U^{be}(x')\,U^{ch}(y)+\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')\\
&-2\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{ce}(y)\,U^{bh}(x')-2\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{be}(y)\,U^{ad}(x)\,U^{ch}(x')\\
&-\frac{1}{2}\big\{\rho^e(x'),\rho^h(y)\big\}\rho^d(x)\,U^{ad}(x)\,U^{bh}(y)\,U^{ce}(x')+\frac{1}{2}\big\{\rho^h(x'),\rho^e(y)\big\}\rho^d(x)\,U^{ad}(x)\,U^{ce}(y)\,U^{bh}(x')\\
&-\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{cd}(x')\,U^{ae}(x)\,U^{bh}(y)+2\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')\\
&+2\,\rho^h(x')\,\rho^e(x)\,\rho^d(y)\,U^{bd}(y)\,U^{ae}(x)\,U^{ch}(x')-\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(y)\\
&+2\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{be}(y)\,U^{ad}(x')\,U^{ch}(x)+2\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{bh}(y)\,U^{ad}(x')\,U^{ce}(x)\\
&+\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(w)\,U^{be}(x')\,U^{ch}(y)-\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(w)\,U^{be}(y)\,U^{ch}(x')\\
&+2\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(y)\,U^{bh}(x')+2\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{be}(y)\,U^{ad}(w)\,U^{ch}(x')\\
&+\frac{1}{2}\big\{\rho^e(x'),\rho^h(y)\big\}\rho^d(x)\,U^{ad}(w)\,U^{bh}(y)\,U^{ce}(x')-\frac{1}{2}\big\{\rho^h(x'),\rho^e(y)\big\}\rho^d(x)\,U^{ad}(w)\,U^{bh}(x')\,U^{ce}(y)\\
&-2\,\rho^h(x')\,\rho^d(y)\,\rho^e(x)\,U^{bd}(y)\,U^{ae}(w)\,U^{ch}(x')-2\,\rho^d(x')\,\rho^h(y)\,\rho^e(x)\,U^{ad}(x')\,U^{ce}(w)\,U^{bh}(y)\\
&+2\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{ce}(y)\,U^{bh}(w')+\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(y)\\
&-2\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{be}(y)\,U^{ch}(x)-2\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(w')\,U^{bh}(y)\,U^{ce}(x)\\
&-2\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(y)\,U^{bh}(w')+2\,\rho^d(x')\,\rho^h(y)\,\rho^e(x)\,U^{ad}(w')\,U^{ce}(w)\,U^{bh}(y)\\
&+\frac{1}{2}\big\{\rho^h(x'),\rho^e(y)\big\}\rho^d(x)\,U^{ad}(x)\,U^{be}(z)\,U^{ch}(w')-\frac{1}{2}\big\{\rho^e(x'),\rho^h(y)\big\}\rho^d(x)\,U^{ad}(x)\,U^{be}(w')\,U^{ch}(z)\\
&+2\,\rho^d(x')\,\rho^h(x)\,\rho^e(y)\,U^{ad}(w')\,U^{be}(z)\,U^{ch}(x)-\frac{1}{2}\big\{\rho^h(x'),\rho^e(y)\big\}\rho^d(x)\,U^{ad}(w)\,U^{be}(z)\,U^{ch}(w')\\
&+\frac{1}{2}\big\{\rho^e(x'),\rho^h(y)\big\}\rho^d(x)\,U^{ad}(w)\,U^{ch}(z)\,U^{be}(w')-\rho^d(x')\big\{\rho^h(x),\rho^e(y)\big\}\,U^{ad}(w')\,U^{be}(z)\,U^{ch}(w)\\
&+\rho^e(x')\big\{\rho^h(x),\rho^d(y)\big\}\,U^{ce}(x')\,U^{bd}(z)\,U^{ah}(w)+\rho^e(x')\big\{\rho^h(x),\rho^d(y)\big\}\,U^{bd}(z)\,U^{ae}(x')\,U^{ch}(w)\\
&-2\,\rho^h(x')\,\rho^d(x)\,\rho^e(y)\,U^{ad}(x)\,U^{be}(z)\,U^{ch}(x')-2\,\rho^e(x')\,\rho^h(x)\,\rho^d(y)\,U^{ae}(x')\,U^{ch}(x)\,U^{bd}(z)\Big]=0\end{aligned}\tag{5.12}$$